# Multidimensional Goodhart

Response Channels, Scorecards, and Residual Shape

# Contents

# 1 The Licensing Problem

## 1.1 The warning and what it does not identify

Goodhart's law is usually introduced through the compressed formulation "When a measure becomes a target, it ceases to be a good measure" [23]. The original macroeconomic warning is sharper about policy use: "Any observed statistical regularity will tend to collapse once pressure is placed upon it for control purposes" [8]. Both formulations describe a real empirical phenomenon: once a measure is used for control, the residual error between proxy and target can acquire direction, support, active constraints, exchange rates, tail behavior, and response dynamics. But none of those shapes is identified by the claim that the proxy became a target.

The same observed score improvement can come from several mechanisms. A fixed pool can be re-ranked so better-scoring units enter the selected set. Continuing units can repair measurement errors with no hidden harm. The same units can game a high-score, high-harm action channel. Or the score can improve because the target itself improved. The aggregate score path does not distinguish those stories. A calculation is licensed only after the response model says what is fixed, what changes, and what hidden value or harm is being measured.

This book treats "multidimensional Goodhart" as a response-modeling framework, not as a single universal theorem. The framework asks what proxy pressure changes and which primitives are declared before importing a bound.

## 1.2 What "multidimensional" actually adds

Let $S$ be a state space. A target map $G : S \to \mathbb{R}^m$ records target-relevant state and a proxy map $P : S \to \mathbb{R}^k$ records measured features. The intended correspondence is $\varphi : \mathbb{R}^m \to \mathbb{R}^k$, with residual $\varepsilon(s) = P(s) - \varphi(G(s))$.

Two gaps matter. The **dimension gap** is the part of target variation the proxy map cannot see, represented by directions such as $\ker \varphi$. The **observation gap** is residual measurement artifact inside the measured domain, represented by $\varepsilon$. A scalar score may hide both gaps at once.

Two response channels matter as much as the geometry. Here $u \in U$ is a declared fixed type: the agent, unit, hospital, model checkpoint, or institution as represented before the policy response being studied. It contains the attributes treated as fixed for the comparison; choices made because of the policy are not part of $u$. The baseline type law is $\nu$. A response kernel $K_{\theta(ds \mid u)}$ is the conditional law of the observed state $s \in S$ for a unit of type $u$ after policy exposure $\theta$. A participation or selection weight $W_{\theta(u)} \geq 0$ changes how much type $u$ appears in the realized population. With these objects,

$$\mu_{\theta(A)} = \frac{\int W_{\theta(u)} K_{\theta(A \mid u)} \nu(du)}{\int W_{\theta(u)} \nu(du)}.$$

Pure selection means $K_\theta = K_0$ for $\nu$-almost every type and policy dependence enters only through $W_\theta$. Intervention means $K_\theta$ changes on a positive-$\nu$ set of fixed types.

Selection and intervention can produce similar marginal score movement, so they do not license the same mathematics.

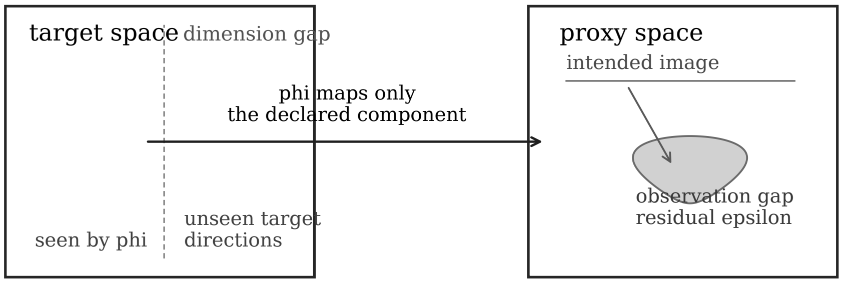


Figure 1: Dimension and observation gaps are different contract fields. The schematic licenses the split in vocabulary, not a quantitative conclusion.

### 1.3 What we tried and what failed

The negative results are not housekeeping; they are the reason the framework has its current shape.

**Claim gallery.**

- **Unconditional dimensional scaling fails.** Hidden harm does not scale with $\dim(\ker \varphi)$ by itself. If hidden coordinates are independent of the selected proxy, thresholding does not move them.
- **"More measured dimensions means more error" has no sign.** Additive and conjunctive aggregation can give opposite comparative statics from the same primitive action channels.
- **Signed aggregate hidden error is unstable.** Positive and negative hidden movements can cancel. A value functional, norm, tail risk, or domain loss has to be declared.
- **Covariance is not a finite-pressure primitive.** With $P = Z$ and $H = Z^2 - 1$, baseline covariance is zero but threshold or Boltzmann selection moves $H$.
- **Absolute continuity is not the intervention boundary.** It is useful for the reweighting theorem, but fixed-type action changes can remain absolutely continuous with respect to the baseline law.
- **The selection/intervention split is not marginally identifiable.** It is relative to a declared type/action representation and needs repeated-type, exposure, action-trace, or structural evidence.
- **The selection bound is not coordinate-free by itself.** It is coordinate-explicit until a value norm or scalar value functional is named.
- **Convex affordability is not welfare.** A score-deficit cost bounds private action affordability, not hidden harm.
- **Minimum-complexity attraction is not generic.** Selection follows baseline tails; intervention follows cost, search, caps, fixed charges, and affordances.

These failures block a single "n-dimensional Goodhart law." What survives is a set of conditional calculations plus a discipline for saying when each calculation applies.

## 1.4 The response-modeling contract

The contract below is not a theorem about what proxy pressure will do. It is a methodological definition: the minimum declaration needed before a proposed Goodhart calculation can be evaluated. Its source is the negative result from the previous section. Marginal score movement does not identify hidden welfare, the type weights $W_\theta$, the response kernels $K_\theta$, the action costs, or the aggregation rule.

Start with a school score. The district announces that funding will depend on a test-score metric. Next year the average score rises. That one fact is compatible with several different stories:

- stronger schools entered the funded pool;
- the same schools taught the underlying material better;
- the same schools drilled the test format while learning did not improve;
- low-scoring students were excluded from the tested population;
- reporting errors were repaired with little hidden harm.

Those stories can have the same score path and require different mathematics. Selection bounds apply to reweighting a fixed population. Intervention bounds need actions, costs, and stakes. Welfare claims need a hidden value or harm model. The response-modeling contract is the rule that says which story is being claimed before any calculation is imported.

The contract has three plain-language jobs:

1. State what the claim wants to conclude.
2. State what response story would make that conclusion meaningful.
3. State what evidence would distinguish that story from nearby alternatives.

Here is the notation used to do that compactly. A type space $U$ is the model's description of the units before the policy response. In the school example, $u \in U$ might be one school with baseline traits such as size, neighborhood, student mix, prior resources, and administrative capacity. The baseline type law $\nu$ says how common those school types are. The observed state $s \in S$ contains what the evaluation later sees, such as test scores, curriculum, student participation, and hidden learning outcomes when they are measured.

The response kernel $K_{\theta(ds \mid u)}$ is the conditional law of the observed state for a fixed type $u$ after policy exposure $\theta$. It is where the model puts what the same school does after seeing the funding rule. The selection weight $W_{\theta(u)} \geq 0$ is where the model puts changes in which fixed types appear in the realized population. With these objects, the induced law from the previous section reads the same way, now with school-score content:

$$\mu_{\theta(A)} = \frac{\int W_{\theta(u)} K_{\theta(A \mid u)} \nu(du)}{\int W_{\theta(u)} \nu(du)}.$$

Pure selection means $K_\theta = K_0$ for $\nu$-almost every type and policy dependence enters only through $W_\theta$. Intervention means $K_\theta$ changes on a positive-$\nu$ set of fixed types. Mixtures are allowed, but they must be named.

A Goodhart claim must therefore declare the primitives that make it a claim rather than a retrospective label:

> **Contract.**
> - **Claimed output:** What are you trying to conclude: score drift, hidden drift, a distribution, a selection/intervention label, an intervention feasibility bound, a welfare comparison, or a response-shape prediction?
> - **Type representation:** What is one fixed unit $u \in U$? For a school score, is $u$ a school before the funding rule, a student, a classroom, or a district? Which attributes are fixed, and which are later choices?
> - **Baseline behavior:** What does a fixed type produce before the policy response? Declare $\nu$ and $K_0(ds \mid u)$: the baseline distribution of observed states conditional on type.
> - **Policy exposure:** What creates pressure, and who sees it: threshold, ranking, scorecard, prize, penalty, benchmark, or feedback signal indexed by $\theta$?
> - **Response channel:** Does $\theta$ only reweight fixed types through $W_{\theta(u)}$, change fixed-type behavior through $K_{\theta(ds \mid u)}$, or both? Selection changes who is represented; intervention changes what a fixed unit does or produces.
> - **Action/search geometry:** If fixed-type behavior changes, what actions are available, what do they cost, what caps or fixed charges exist, what search process is used, and what stakes $V$ make response worthwhile?
> - **Proxy/target relation:** What proxy $P$ is optimized, what target $G$ or hidden quantity $H$ is protected, what relation $P \approx \varphi(G)$ is intended, and where are the dimension and observation gaps?
> - **Aggregation:** If there are multiple proxy components, how are they combined: additive weights, thresholds, conjunctive gates, Pareto rules, lexicographic rules, or an institution-specific formula?
> - **Hidden value or harm:** What makes a response good or bad beyond the proxy: hidden coordinates, scalar value vector, norm, loss, or harm rates $h_j$?
> - **Evidence standard and falsifier:** What observations would distinguish the claimed story from nearby alternatives, and what observation would make this contract the wrong one?

The first row is load-bearing. The contract is not a single formula with one fixed output. It is an adequacy test for a proposed output. A scalar hidden drift claim needs less information than a full response kernel. A distribution over $k$ discrete states has $k-1$ degrees of freedom. A hidden drift vector in $\mathbb{R}^m$ needs $m$ value-relevant coordinates. A response-kernel claim can be infinite-dimensional unless the model restricts it. A selection/intervention label has a small output alphabet, but it is causal: it cannot usually be read from the marginal score path.

The type representation is part of the empirical claim, not notation to hide inside the model. If $U$ includes each school's whole future response plan, it can make every intervention look like selection over richer types. If $U$ is too coarse, stable heterogeneity can look like a kernel change. The contract therefore has to defend why the chosen $u$ is fixed for the comparison and why omitted variation belongs in $K_\theta$, $W_\theta$, or the action model. The induced marginal law $\mu_\theta$ usually cannot distinguish those choices by itself.

After the output is named, the contract should pass a small information accounting check:

> **Contract adequacy.**
> 1. State the output object and its degrees of freedom or identifying data.
> 2. List the raw primitives the claim uses: type law, kernels, weights, proxy and target maps, aggregation, action costs, stakes, hidden value, and evidence.
> 3. Subtract constraints, such as probability normalization, declared equality $K_\theta = K_0$, additive aggregation, convex cost, or fixed deficit.
> 4. Subtract redundancies, such as the arbitrary scale of $W_\theta$ or coordinate rescalings that leave the represented situation unchanged.
> 5. Check units: costs must be comparable to stakes, proxy gains to score deficits, and hidden harm rates to the declared value units.
> 6. Check rank or relevance: independent changes in the declared inputs should affect the claimed output, unless the redundancy has been stated.
> 7. Check invariance: the conclusion should not depend on arbitrary naming, coordinate splitting, units, or post-hoc enrichment/coarsening of $U$.

In shorthand, a contract is adequate only if

$$\text{true input information} = \text{raw variables} - \text{constraints} - \text{redundancies}$$

is enough to identify the claimed output up to the intended invariances. If two meaningfully different response stories still satisfy the same declared inputs, the contract has not licensed that output. It may still license a weaker output: for example, an aggregate score path may license a monitoring trigger while not licensing a welfare verdict or a selection/intervention classification.

The contract is demanding by design. It prevents the framework from inferring welfare, hidden target movement, or a response channel from marginal score movement alone.

# 2 Licensed Calculations

## 2.1 Selection channel

Before the theorem can be used, the contract must supply a baseline law, a pure selection channel, hidden coordinates or a value metric, and enough integrable moments. Pure selection means $K_\theta = K_0$ and the policy changes only the weights, with the induced law absolutely continuous with respect to baseline, $\mu_\theta \ll \mu_0$. Write $L = d\frac{\mu_\theta}{d}\mu_0$ and $\delta = \|L - 1\|_{L^2(\mu_0)}$, and call $B_{H(\theta)} = \mathbb{E}_{\mu_\theta}[H] - \mathbb{E}_{\mu_0}[H]$ the selection drift of the hidden vector $H$.

**T1. Coordinate-explicit selection bound.** For hidden coordinates $H_i$ with finite second moments and baseline standard deviations $s_i$,

$$\left|\mathbb{E}_{\mu_\theta}[H_i] - \mathbb{E}_{\mu_0}[H_i]\right| \le \delta s_i$$

for each coordinate. After a Euclidean coordinate metric is declared,

$$\left\|B_{H(\theta)}\right\|_2 \le \delta \|s\|_2$$

.

**T2. Value-weighted/operator selection bound.** For a declared scalar value direction $v$ and covariance $\Sigma_H$,

$$\left|v \cdot B_{H(\theta)}\right| \le \delta \sqrt{v^T \Sigma_H v}$$

. For a declared norm,

$$\left\|B_{H(\theta)}\right\|_V \le \delta \sup_{\|v\|_{V,*} \le 1} \sqrt{v^T \Sigma_H v}$$

.

These are Hilbert-space Cauchy–Schwarz bounds. They license drift envelopes for pure reweighting. They do not identify the hidden coordinates, the value weights, or the welfare object. They do not apply to fixed-type response changes.

Covariance belongs here only as a local velocity. Along a valid exponential tilt, the derivative at zero pressure is a covariance. At finite pressure, the path response, tail shape, and moment-generating domain matter. That is why the zero-covariance example $H = Z^2 - 1$ survives as a warning.

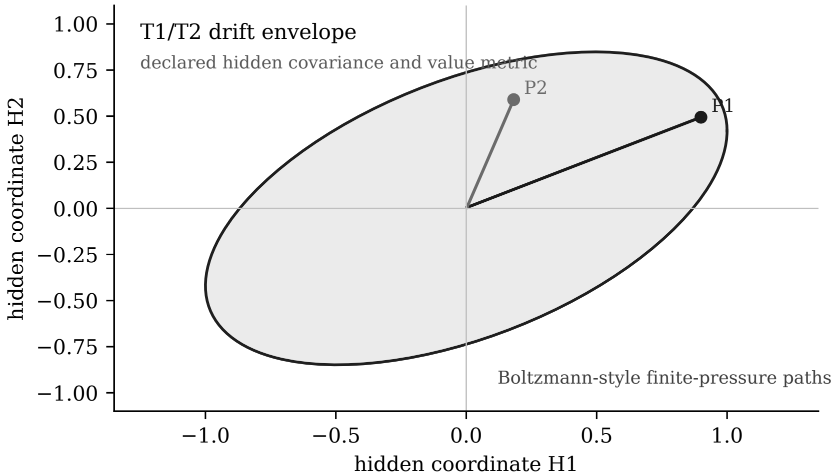


Figure 2: T1/T2 give a declared hidden-space envelope. Boltzmann-style pressure paths are trajectories inside that declared geometry, not replacements for the finite-pressure bound.

## 2.2 Intervention channel

Before an intervention calculation can be used, the contract must supply a fixed-type action model: actions $a$, private cost $c(a)$, proxy gain $w \cdot a$, score deficit $d$, and stakes $V$. The calculation measures the affordability of proxy movement under that model.

> **T3. Quadratic Stackelberg wedge.** In the one-dimensional noiseless threshold toy with quality $Q$, action $a \geq 0$, private cost $\frac{a^2}{2\kappa}$, threshold $t$, and stakes $V$, gaming by a below-threshold unit is privately worthwhile exactly when $t - Q \leq \sqrt{2\kappa V}$.

The wedge $\sqrt{2\kappa V}$ is a signature of this quadratic toy. It is not a universal intervention law and should not be read as an RLHF, benchmark, or organizational theorem unless the action, cost, stakes, and pass condition have been declared.

> **T4. Convex score-deficit budget.** With finite-dimensional action space, closed proper convex cost $c$, linear proxy gain $w \cdot a$, and regularity for convex duality, define $m(d) = \inf_a \{c(a) : w \cdot a \geq d\}$. Then
>
> $$m(d) = \sup_{\lambda \geq 0} [\lambda d - c^*(\lambda w)]$$
>
> . Gaming under stakes $V$ is feasible exactly when $m(d) \leq V$ in this declared private-cost model.

Cost minimization and hidden-welfare assessment answer different questions. For example, if two proxy channels have equal private cost and equal score weight but harm rates $(h, 0)$, cost minimization splits effort while hidden harm grows with $h$. The budget licenses private affordability; welfare requires a hidden harm functional.

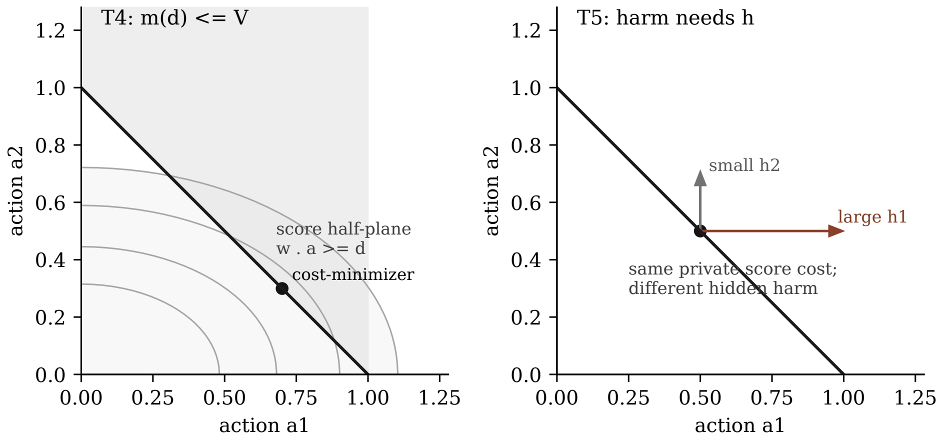


Figure 3: T4 locates the private cost-minimizing action for a score deficit. T5 then requires hidden harm exchange rates before that action can be interpreted as welfare movement.

## 2.3 Multidimensional scorecards — the keeper

The scorecard calculation needs more than "more metrics." It needs an additive score, measured channel set $M$, weights $w_j$, separable quadratic costs $\frac{a_j^2}{2\kappa_j}$, fixed score deficit $d$, and linear hidden harm $\sum_{j\in M} h_j a_j$.

> **T5. Additive exchange-rate iff.** The fixed-deficit per-agent hidden harm of the cost-minimizing action is
>
> $$H_{M(d)} = d\frac{\sum_{j\in M} h_j \kappa_j w_j}{\sum_{j\in M} \kappa_j w_j^2}.$$
>
> Fixed-deficit harm is conserved across compared active measured sets exactly when $h_j = cw_j$ on those active channels.

This is the most exportable scorecard result. It says what must be checked before saying whether adding or removing a measured channel helped, hurt, or only re-routed harm: the hidden harm per score unit has to be declared or estimated. Holmstrom–Milgrom multitasking is the closest economics precedent, but the contract here makes the hidden-harm exchange rate explicit.

The result is fixed-deficit and per-agent. Population entry is separate: lowering the private cost of reaching the score can recruit more units into gaming even when each fixed-deficit gamer has conserved harm. Conjunctive aggregation is separate too; requiring every measured component to clear a bar can make harm grow with the number of components.

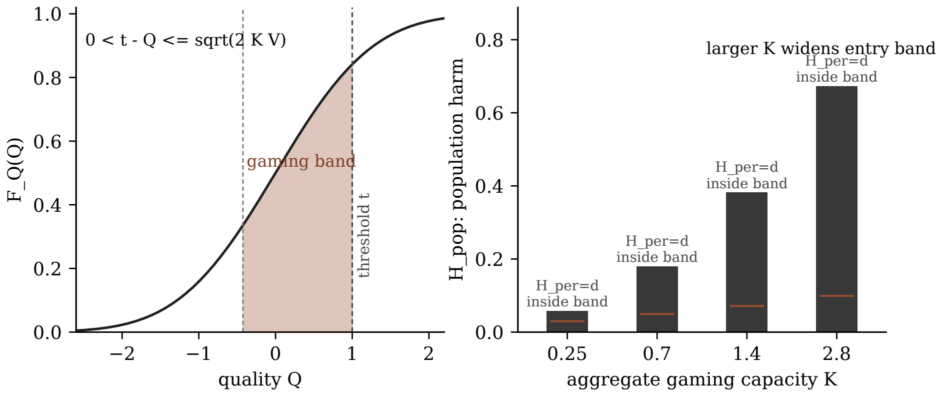


Figure 4: The gaming band separates fixed-deficit per-gamer harm $H_{\text{per}}$ from population harm $H_{\text{pop}}$. More capacity widens entry without by itself changing the per-gamer exchange-rate formula.

## 2.4 Adaptive hardening — narrow but real

The hardening result is narrow by design. The contract is fixed finite measured set $M$, fixed deficit $d$, fixed stakes $V$, fixed weights, additive proxy gain, separable quadratic costs, deterministic observation, and monotone hardening of capacities $\kappa_{j,t}$.

> **T6. Deterministic adaptive-hardening capacity boundary.** Let
>
> $$S_{t(M)} = \sum_{j \in M} \kappa_{j,t} w_j^2$$
>
> and $T = \frac{d^2}{2V}$. At time $t$, fixed-deficit gaming is feasible exactly when $S_{t(M)} \geq T$. Hardening reaches no-gaming exactly when $S_{t(M)} < T$. A progress-aware largest-action multiplicative rule terminates in finite time when channels are finite, positive-weight, and floor capacity satisfies $S_{\text{floor}(M)} < T$.

Nothing broader is licensed. The theorem does not cover stochastic observation, arbitrary hardening rules, changing measured sets, changing deficits, changing stakes, nonconvex costs, shared bottlenecks, cycles, welfare comparison, or policy optimality. The immediate anti-transfer is noisy observation: once the update rule can chase noise, the deterministic threshold is no longer a stopping theorem.

## 2.5 Response shape, conditionally

The project did not prove a generic residual-shape law. What survives is a conditional taxonomy:

- Quadratic costs select the minimum-cost direction, often proportional to $Cw$ in a smooth unconstrained model.
- Fixed charges create entry thresholds and active-set comparisons.
- Caps produce spillover only after a channel saturates.
- Low-rank affordances constrain response to an image, but do not choose a basis-invariant "simple" direction by themselves.
- Search-prior claims require a coding language or search process fixed before the failure is observed.

These are not theorem transfers between domains. They are ways to turn a response-shape conjecture into a declared model with falsifiers.

# 3 Across Disciplines: Primitive Attribution

## 3.1 Genealogy

Goodhart, Campbell, Strathern, and Manheim–Garrabrant are genealogy, not proof sources for the calculations above. Goodhart's original macroeconomic warning concerns policy-contaminated regularities [8]; the Lucas critique is its companion precedent, arguing that policy change invalidates the estimated relations policy relies on [14]. Campbell's warning concerns social indicators under decision pressure [3]. Strathern's formulation supplies the familiar compressed warning [23]. Manheim and Garrabrant provide a useful cause taxonomy [16].

This book does not replace that genealogy with a grander taxonomy. It uses the genealogy to ask which primitives are present: selection, response kernel, action cost, proxy/target separation, aggregation, hidden value, and evidence standard.

## 3.2 Formal analogues, primitive by primitive

The closest formal analogues supply some primitives and omit others. El-Mhamdi–Hoang and Majka–El-Mhamdi are scalar tail-conditioned selection anchors [6, 15]. Hardt et al. strategic classification supplies costly feature-change intervention [9]. Perdomo et al. performative prediction supplies deployment-induced response kernels [18]. Skalse et al. supply proxy/target and RL-specific response-kernel tools [13, 21]. Holmstrom–Milgrom supplies the closest multitask incentive analogue for aggregation and action costs [12]. Cawley–Talbot, reusable holdout, leaderboard work, and optimizer's curse supply selection and evidence standards [2, 4, 5, 19, 20, 22]. Pan and Gao provide empirical response-geometry candidates, not plug-in parameters [7, 17].

## 3.3 What each discipline contributes vs. omits

| **Source family** | **Primitives supplied** | **Primitives omitted** | **Licensed transfer** |
|---|---|---|---|
| Tail-conditioned Goodhart | Selection, scalar proxy/target, tail assumptions. | Vector aggregation, intervention, hidden welfare. | Sharp scalar asymptotics where the tail contract holds; otherwise only a reweighting analogy. |
| Strategic classification and performative prediction | Response kernels, costly action or endogenous deployment. | General hidden value, scorecard exchange rates, marginal identification. | Intervention framing after action/cost or deployment response is declared. |

| Source family | Primitives supplied | Primitives omitted | Licensed transfer |
|---|---|---|---|
| RL reward hacking | Proxy/target separation, MDP occupancy geometry, optimization pressure. | Non-MDP domains, empirical welfare, generic cost parameter. | RL-specific response calculations only inside the MDP/occupancy contract. |
| Multitask incentives | Aggregation and multidimensional effort/cost. | The book's value-metric contract, non-LEN settings unless separately shown. | External precedent for the scorecard exchange-rate question, not a proof of this book's theorem. |
| Adaptive holdout and leaderboard hygiene | Selection/evidence standards for repeated benchmark use. | Fixed-type fine-tuning, contamination, welfare, action cost. | Evidence discipline for selection over fixed candidates and public/private score interpretation. |
| Reward overoptimization empirics | Candidate search and optimization-pressure variables. | A defended mapping from model size, KL, data access, or prize to $\kappa$ or $V$. | Hypothesis generators for response geometry; no automatic theorem transfer. |

## 3.4 Reduction, not unification

No single equation unifies these literatures. The framework's claim is narrower: each formalism fills different fields of the response-modeling contract. The contract says which primitive is being borrowed and which transfer fails. A source about adaptive holdouts does not identify hidden welfare. A source about costly feature change does not identify value weights. A source about multitask contracts does not make "more metrics" good or bad in a new application.

# 4 Cases as Stress Tests

## 4.1 MMLU

MMLU is useful because it is not a single mechanism [10]. A public benchmark score can induce fixed-checkpoint model selection, repeated adaptive testing, prompt search, finetuning on benchmark-like data, contamination, reward-model overoptimization, or genuine capability transfer.

The contract separates these mechanisms. Fixed-checkpoint comparison is a selection problem: candidate checkpoints are types, the score is a proxy, and the hidden target is performance away from the benchmark. Finetuning, prompt search, contamination, or reward optimization is intervention/search only after an action/search geometry is declared. MMLU alone supplies neither $\kappa$ nor stakes $V$, and it does not supply hidden welfare. T3/T4 apply only after a cost model and score deficit are declared. T5 applies only after benchmark components are modeled as additive channels with harm exchange rates.

The stress test is whether the evidence can distinguish fixed-candidate selection from post-exposure adaptation. Necessary observations include benchmark access, training data lineage, prompt/search budget, repeated-query history, transfer to non-MMLU tasks, and failure modes that move when the benchmark format changes.

## 4.2 Hospital readmissions

For a readmission scorecard, score improvement is compatible with hospitals leaving the comparison pool, coding changes, observation-status changes, delayed admissions, patient avoidance, better follow-up care, or mixtures.

The auditable primitives are concrete: effective score weights $w$, stakes $V$, response ease $\kappa$ for coding, discharge timing, follow-up, and patient selection channels, hidden harm rates $h$, and the signal adequacy of the measured components. If these are unavailable, the design consequence is not "assume the score is bad." It is: do not use the toy diagnostic, collect action traces, monitor hidden patient outcomes where possible, pilot or lower leverage, and mark the missing primitive.

This is an evidence contract, not policy advice. The framework can say that aggregate readmission movement is insufficient to credit patient-welfare improvement. It cannot rank hospital policies without the clinical and value model the contract explicitly requires.

## 4.3 Scientific metrics

Publication counts, citations, grants, venue prestige, and rankings are already covered by responsible-metrics warnings such as DORA [1] and the Leiden Manifesto [11]. The framework does not claim novelty for the warning that crude metrics can distort science.

The stress test is discriminator value. A scientific scorecard application must separate selection over researchers, labs, fields, and institutions from fixed-researcher response. It must distinguish harmful proxy manufacture from harmless metadata/

discoverability repair and from genuine improvements in methods, robustness, replication value, useful synthesis, or long-run uptake.

The contract refuses broad research-value inference from score movement. If an institution cannot declare the target, aggregation, action traces, and hidden value model before interpreting a score rise, the framework returns "no licensed hidden-value claim."

# 5 Practical Implications

## 5.1 What to do before crediting score movement

Wentworth's practical warning is that experiments often measure something other than what the experimenter thinks they are measuring: an unexpected confounder, a secondary channel, a sampling artifact, or an operational detail that happens to track the named quantity [24]. The proposed repair is not to stare harder at the headline metric. It is to measure many auxiliary traces so the unexpected channel has somewhere to show up.

This book agrees with the first half and adds a second step. Auxiliary traces are necessary for discovery, but they are not sufficient for interpretation. They must feed a declared response contract. The combined rule is: collect the firehose for discovery, then use the contract for claim licensing. Without the firehose, the analyst misses the confounder. Without the contract, the analyst has many traces and still no disciplined statement of which score movement counts as selection, fixed-type response, proxy repair, harmful gaming, real improvement, or a mixture.

The practical workflow is therefore not to ask first, "did the score improve?" Ask instead: what changed, who changed, by what channel, at what cost, with what hidden target evidence, and what observation would make this interpretation fail? A score rise is an observation to be explained, not a conclusion to be credited.

| **Audit field** | **Entry before crediting score movement** |
|---|---|
| Observed score movement | Magnitude, timing, affected units, score components, and whether the movement is marginal, thresholded, ranked, or aggregate. |
| Candidate mechanisms | Selection, fixed-type response, proxy repair, harmful gaming, real improvement, or an explicit mixture. |
| Required discriminator traces | Repeated-type evidence, exposure timing, action logs, search or query budget, component-level score changes, participation changes, and off-score outcomes that separate nearby mechanisms. |
| Hidden value/harm evidence | Declared target coordinates, value weights, harm rates, residual measures, or domain outcomes; if absent, mark the hidden-value claim unlicensed. |
| Missing primitives and operational consequence | The undeclared type space, baseline law, response channel, action cost, aggregation rule, or hidden value model, plus the resulting consequence: collect traces, lower leverage, pilot, or withhold the claim. |
| Licensed claim | The narrow statement supported by the declared contract: for example, pure-selection drift envelope, private affordability under a cost model, exchange-rate diagnostic, or no hidden-value conclusion. |

| Audit field | Entry before crediting score movement |
|---|---|
| Blocked claim | The stronger statement not supported: welfare improvement, target improvement, policy optimality, channel identification, or generic anti-metric advice. |
| Contract-failure condition | A concrete observation that would defeat the interpretation, such as action traces inconsistent with the named channel, hidden outcomes moving opposite the claimed target, or theorem conditions failing. |

This sheet is intentionally smaller than the full application template. Its job is to slow down the common interpretive jump. If the observed movement is a readmission decline, a benchmark rise, a citation increase, or a ranking gain, the first audit question is the same: what response channel could have produced this movement? The second is which traces would discriminate that channel from its nearest rivals. The third is which hidden value or harm evidence is present, and which claim remains blocked if it is absent.

The conclusion can be positive, but it has to be narrow. "The score improved and auxiliary outcomes moved in the declared target direction under stable exposure" is a different claim from "the policy improved welfare." "The action logs fit a low-cost proxy-repair story" is a different claim from "the metric is safe." "The selection envelope bounds hidden coordinate drift under the stated law" is a different claim from "selection is harmless." Practical use of the framework is mostly this discipline of replacing a large conclusion with the smaller one the evidence actually licenses.

# 6 What the Framework Refuses to Do

## 6.1 Anti-applications

The framework is least useful when its primitives cannot be stably declared. Concrete anti-applications include:

- A rapidly shifting online platform where type space, measured set, policy exposure, and available actions all change between measurements.
- A black-box institutional ranking where effective weights and thresholds are unobservable and participants cannot tell which action channel is being rewarded.
- A high-stakes public decision where stakeholders demand a welfare verdict but no defensible hidden value model exists.

In these cases, the contract returns "no verdict." That is not a hedge; it is the framework refusing to convert score movement into causal mechanism or welfare language without the primitives that would make the claim testable.

## 6.2 Falsifiers

The contract is falsifiable as a modeling discipline, but the failure point has to be located honestly. The theorems themselves cannot be empirically violated while their hypotheses hold; they are proved. What can fail is the claim the contract makes about a domain: that independently audited declarations of channel, costs, stakes, and harm rates track the response mechanism well enough for the licensed calculation to predict response shape or direction. A concrete falsifier is therefore a domain where the primitives are declared and audited in advance, response channel and action traces are observable enough to distinguish nearby mechanisms, and the licensed calculation still systematically predicts wrongly while a simpler score-only rule predicts correctly.

Examples include:

- A setting independently audited as pure selection, with the declared moment and absolute-continuity conditions checked, where hidden drift repeatedly exceeds the $\delta s_i$ envelope. The bound cannot fail under its hypotheses, so repeated excess drift would show that the audit and the contract's response-channel field failed to detect a fixed-type response channel.
- A hardening setting audited as having stable $M$, $d$, $V$, weights, separable quadratic costs, and deterministic observation, where gaming remains feasible after $S_{t(M)} < \frac{d^2}{2V}$ or stops while $S_{t(M)} \geq \frac{d^2}{2V}$ — showing the audited capacities or observation model were not the ones the agents faced.
- An additive scorecard with defended $w_j$, $\kappa_j$, and $h_j$ where fixed-deficit per-agent harm is conserved across active measured sets even though the exchange-rate condition $h_j = cw_j$ fails, or moves even though the condition holds — showing the declared channels, costs, or harm rates were not the ones generating the response.

Such failures, if they survived audit, would not call for prose repair. They would mean the contract fields do not track the response mechanisms they were designed to track — that disciplined declaration does not buy predictive power. That is the falsifiable content of the framework.

# 7 Open Agenda

## 7.1 The residual-shape conjecture

The signature open problem is the residual-shape conjecture: when does repeated proxy repair drive hidden failure toward a predictable residual shape, such as a low-complexity attractor? The current answer is negative unless a mechanism is named first.

Resolving it requires at least five declarations: a response mechanism, a complexity or shape functional fixed before inspection, a policy-update rule, a composition rule for repeated repair, and a failure condition. It would count as progress to prove a sparse, low-rank, or low-description-length attractor inside one of those contracts. It would not count to observe a simple-looking failure after the fact and relabel it as the attractor.

## 7.2 Composition, identification, and information

Three structural gaps remain.

First, channels compose. Real cases mix selection, fixed-type response, proxy repair, and real improvement. The current contract names mixtures but does not give a general calculus for them.

Second, primitives are not automatically identifiable. Marginal score movement does not identify $W_\theta$ versus $K_\theta$, hidden harm, action costs, or aggregation. The framework needs identification toys and evidence thresholds for when a primitive can be treated as declared rather than guessed.

Third, the $L^2$ selection bound may not be the portable final form. An information-theoretic restatement could travel further if it preserves the distinction between coordinate-explicit drift, declared value metrics, and finite-pressure path behavior.

## 7.3 Toolkit gap

A practitioner should not have to re-derive the contract every time. The missing toolkit has three parts: a primitive-elicitation protocol, worked exchange-rate audits in real scorecards, and small identification examples that show what observations distinguish selection, intervention, proxy repair, and real improvement.

Until those tools exist, the framework is best read as a claim-license discipline and a theorem inventory. It tells a reader what would have to be true before a Goodhart calculation travels.

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